Lee Yeong Kim

Postdoctoral Researcher, University of Vienna

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Research Interests

My PhD thesis focuses on matter-wave scattering (4He and D2) with thermal-energy atoms incident on surfaces at extremely small incidence angles. This approach enables the study of the atom-surface dispersion potential by accessing a kinetic-energy range of 2-100 neV. We demonstrated fundamental optical principles – Babinet's principle and the reciprocity theorem – in the context of matter waves, and discovered a new non-destructive reflection mechanism, multiple edge-diffraction reflection.

In addition, I actively contributed to molecular deflection experiments using a counter-propagating, non-resonant laser pulse and a pulsed molecular beam. The deflected molecules were ionized and detected using velocity map imaging mass spectrometry. I mainly focused on molecular trajectory simulations, considering the rotational-state-dependent polarization of molecules in laser fields.

Motivated by these experiences, I am interested in quantum optics, quantum phenomena on solid surfaces, dispersive interactions between materials, and AMO physics such as laser cooling, trapping, and precision spectroscopy.

Education

Combined MS/PhD — **Ulsan National Institute of Science and Technology (UNIST)**, Physics — Ulsan, Korea
Advisor: Prof. Bum Suk Zhao. Dissertation: "Study on matter-wave optics under grazing incidence condition with solid surfaces." — Mar 2016 – Aug 2022
- Grade: 4.02/4.3

BS — **Ulsan National Institute of Science and Technology (UNIST)**, Physics (Minor: Energy Engineering) — Ulsan, Korea
Mar 2011 – Feb 2016
- Grade: 3.60/4.3

Academic Experience

University of Vienna, Postdoctoral Researcher — Vienna, Austria
Research advisor: Prof. Markus Arndt — Jan 2026 – present

Career transition period (spouse's postdoctoral stay), Seeking academic opportunities to continue my research career — Zürich, Switzerland
July 2023 – Dec 2025
Took an unplanned career break of approximately two years to accompany my spouse, while remaining actively engaged in research: two publications in 2024 and participation in the MOLEC 2024 and ISMB 2025 conferences.

UNIST, Postdoctoral Researcher — Ulsan, Korea
Research advisor: Prof. Bum Suk Zhao — Sept 2022 – June 2023

Fritz-Haber-Institut der Max-Planck-Gesellschaft, Visiting Student — Berlin, Germany
Research advisors: Dr. Wieland Schöllkopf and Prof. Bum Suk Zhao — Jan 2014 – Dec 2019
- Seven visits: 4 months (2014, 2017), 2 months (2015, 2016), and 10 months (2019).

UNIST, Undergraduate Research Assistant — Ulsan, Korea
Research advisor: Prof. Bum Suk Zhao — July 2013 – Feb 2016

Funding and Fellowships

quantA Postdoctoral Fellowship: Empowering Women in Quantum Science 2026 – 2027
quantA

Global Ph.D. Fellowship 2017 – 2021
The Korean Government – acceptance rate 7.6:1

Global Internship Program Scholarship July 2014 – Nov 2014
The Korean Government

Awards and Honors

Seal of Excellence, Marie Skłodowska-Curie Actions Postdoctoral Fellowship Feb 2026
European Commission – proposal score 95.8/100

UNIST Best Research Award Feb 2023
UNIST – top 5 excellent graduates

Best Poster Presentation Award Aug 2021
The Korean Physical Society

Best Poster Presentation Award Apr 2021
The Korean Physical Society

Best Oral Presentation Award Apr 2016
The Korean Physical Society

Ulsan Mayor Award Feb 2016
Ulsan – top 5 excellent graduates

Publications

Comprehensive peak-width analysis in matter-wave diffraction under grazing incidence conditions 2024
Lee Yeong Kim, Do Won Kang, Jong Chan Lee, Eunmi Chae, Wieland Schöllkopf, Bum Suk Zhao
10.1103/PhysRevA.110.013313 (Phys. Rev. A 110, 013313)

Diffractive mirrors for neutral-atom matter-wave optics 2024
Lee Yeong Kim, Do Won Kang, Sanghwan Park, Seongyeop Lim, Jangwoo Kim, Wieland Schöllkopf, Bum Suk Zhao
10.1039/D3FD00155E (Farad. Disc. 251, 160)

Enhanced elastic scattering of He2 and He3 from solids by multiple-edge diffraction 2022
Lee Yeong Kim, Sanghwan Park, Chang Young Lee, Wieland Schöllkopf, Bum Suk Zhao
10.1039/D2CP02641D (Phys. Chem. Chem. Phys. 24, 21593 (Front cover))

Experimental test of Babinet's Principle in matter-wave diffraction 2021
Lee Yeong Kim, Ju Hyeon Lee, Yun-Tae Kim, Sanghwan Park, Chang Young Lee, Wieland Schöllkopf, Bum Suk Zhao
10.1039/D0CP05694D (Phys. Chem. Chem. Phys. 23, 8030)

Scattering of adiabatically aligned molecules by nonresonant optical standing waves 2020
Lee Yeong Kim, Byung Gwun Jin, Tae Woo Kim, Ju Hyeon Lee, Bum Suk Zhao
10.1126/sciadv.aaz0682 (Sci. Adv. 6, eaaz0682)

Matter-wave diffraction from a periodic array of half planes 2019
Ju Hyeon Lee, **Lee Yeong Kim**, Yun-Tae Kim, Chang Young Lee, Wieland Schöllkopf, Bum Suk Zhao
10.1103/PhysRevLett.122.040401 (Phys. Rev. Lett. 122, 040401 (Editor's Suggestion))

Universal diffraction of atomic and molecular matterwaves: A comparison of He and D2 quantum reflected from a grating 2016
Weiqing Zhang, Ju Hyeon Lee, Hye Ah Kim, Byung Gwun Jin, Bong Jun Kim, **Lee Yeong Kim**, Bum Suk Zhao, Wieland Schöllkopf
10.1002/cphc.201600965 (Chem. Phys. Chem. 17, 3670)

Strong optical dipole force exerted on molecules having low rotational temperature 2016
Xing Nan Sun, Byung Gwun Jin, **Lee Yeong Kim**, Bong Jun Kim, Bum Suk Zhao
10.1002/cphc.201600838 (Chem. Phys. Chem. 17, 3701)

Effect of rotational-state-dependent molecular alignment on the optical dipole force 2016
Lee Yeong Kim, Ju Hyeon Lee, Hye Ah Kim, Sang Kyu Kwak, Bretislav Friedrich, Bum Suk Zhao
10.1103/PhysRevA.94.013428 (Phys. Rev. A 94, 013428)

Rotational-state-dependent dispersion of molecules by pulsed optical standing waves 2015
Xing Nan Sun, **Lee Yeong Kim**, Bum Suk Zhao, Doo Soo Chung
10.1103/PhysRevLett.115.223001 (Phys. Rev. Lett. 115, 223001)

Skills

Laboratory Techniques: Precise scattering experiments with matter-waves (4He, D2) under high vacuum (10^-6 to 10^-12 mbar); vacuum systems (diffusion, rotary, NEG, and turbo pumps); operation and troubleshooting of nanosecond IR (Nd:YAG: Surelite, Powerlite) and visible dye (Sirah) lasers; Time-of-Flight (TOF) mass spectrometry, Velocity Map Ion Imaging (VMI), Resonance Enhanced Multiphoton Ionization (REMPI); Atomic Force Microscopy (Dimension ICON, NX10)

Computational Skills: Molecular trajectory calculations with the Monte-Carlo method, solving the Schrödinger equation; experimental automation using LabVIEW; converted and upgraded simulations between C, Python, Mathematica, C++, Matlab, and LabVIEW

Computer Programs: C++, Matlab, LabVIEW, LaTeX, Origin, Adobe Illustrator

Research Skills: Experimental design and execution, data analysis and interpretation, technical/procedure documentation, scientific writing and presentation

Languages: Korean (native), English (upper intermediate), German (intermediate, B1 TELC certificate)

Leadership: Student representative, Department of Electrical and Electronic Engineering, UNIST (2012) – liaison between students and faculty, organizing departmental meetings and student activities

Supervising and Mentoring

Mentoring and Teaching
- Provided guidance to 4 graduate and 3 undergraduate students in research projects
- Served as teaching assistant (TA) 5 times for experimental physics laboratory courses
- Designed experiments and assisted as TA twice for physical chemistry laboratory courses

Conference Activities

Conference on Quantum Fluid Clusters, QFC 2026 Austria
Poster June 2026

COST COSY WG4 Meeting Austria
Oral May 2026

2025 International Symposium on Molecular Beams (ISMB) Switzerland
Poster July 2025

24th European Conference on the Dynamics of Molecular Systems Poland
Poster July 2024

2021 Korean Physical Society Fall Meeting Korea
Poster Oct 2021

Invited talk, 2021 International Symposium on Molecular Beams (ISMB) Online
Oral July 2021

2021 Korean Physical Society Spring Meeting Korea
Poster Apr 2021

The 5th Asian Workshop on Molecular Spectroscopy (AWMS) Japan
Oral Mar 2021

2020 Society of Global Ph.D. Fellows Annual Conference Korea
Oral Nov 2020

2020 Korean Physical Society Fall Meeting Korea
Oral Nov 2020

2019 Society of Global Ph.D. Fellows Annual Conference Korea
Oral Nov 2019

2019 Korean Physical Society Fall Meeting Korea
Poster Oct 2019

2018 Frontiers of Matter Wave Optics (FOMO) Greece
Poster Sept 2018

2017 Korean-French Joint Symposium Korea
Poster Sept 2017

Hot topic, 2017 International Symposium on Molecular Beams (ISMB) Netherlands
Oral June 2017

2017 Korean Physical Society Spring Meeting Korea
Oral Apr 2017

2016 Korean Physical Society Fall Meeting Korea
Poster Oct 2016

Scattering of Atoms and Molecules from Surfaces (SAMS) Norway
Poster Aug 2016

2016 Korean Physical Society Spring Meeting Korea
Oral Apr 2016

The 13th Symposium on Gas-phase Chemical Dynamics & Spectroscopy Korea
Oral Feb 2016

The 12th Symposium on Gas-phase Chemical Dynamics & Spectroscopy Korea
Oral Oct 2015

2015 Korean Physical Society Spring Meeting Korea
Oral Apr 2015

2015 Korean Chemical Society Spring Meeting Korea
Oral Apr 2015

2015 Symposium on Molecular Beams (ISMB) Spain
Poster July 2015